

0.1 Standard Model of Particle Physics

The Standard Model (SM) of particle physics is a framework which contains our current understanding of the matter and three of the four forces found in the universe. Within the SM there are 12 elementary fermions, 5 vector bosons which mediate the three fundamental forces and a scalar boson responsible for particle masses (Figure 1). It is important to note that the SM is not a complete model of the universe. A great example of this is the gravitational force which is not described by the theory, however at the small scales of particle physics this force is neglected due to its insignificant strength when compared to the other three forces (strong, electromagnetic and weak).

Quarks	u Up	c Charm	t Top	g Gluon	H Higgs
	d Down	s Strange	b Bottom	γ Photon	Gauge Bosons
Leptons	e Electron	μ Muon	τ Tau	Z Z Boson	
	ν_e Electron Neutrino	ν_μ Muon Neutrino	ν_τ Tau Neutrino	W W Boson	

Figure 1: The Standard Model of particle physics with the 12 fermions shown in purple and green, the five vector bosons shown in red and the Higgs scalar boson shown in yellow.

The SM comprises of a quantum field theory (QFT) and is based on the non-Abelian gauge symmetry groups $SU(3) \times SU(2) \times U(1)$. The $SU(3)$ governs the strong interactions and the $SU(2) \times U(1)$ the electroweak interactions. Each group contains their respective gauge bosons: gluon for the strong and W,Z and photon for the electroweak. The $SU(3)$ is an exact gauge symmetry where as the electroweak symmetry is spontaneously broken by the Higgs field, resulting in the massive electroweak bosons ($W^\pm$ and $Z^\pm$). Due to this broken symmetry which gives rise to massive

quarks, the strong and weak eigenstates diverge. The weak eigenstates (d' , s' , b') of down, strange and bottom quarks which are associated to the weak interaction are related to the mass eigenstates (d , s , b) through the Cabibbo-Kobayashi-Maskawa (CKM) matrix V_{CKM} [1, 2].

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = V_{\text{CKM}} \begin{pmatrix} d \\ s \\ b \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix} \quad (0.1)$$

Bibliography

- [1] N. Cabibbo, “Unitary symmetry and leptonic decays,” *Phys. Rev. Lett.*, vol. 10, pp. 531–533, Jun 1963.
- [2] M. Kobayashi and T. Maskawa, “Cp-violation in the renormalizable theory of weak interaction,” *Progress of Theoretical Physics*, vol. 49, pp. 652–657, 02 1973.